

Kumar C

DATA SCIENTIST

kumaragri448@gmail.com — +91 83448 32394 / 63820 42482 — Dharmapuri, Tamil Nadu, India — github.com/itzkumarc

Profile

Data Scientist with 5+ years of experience building and deploying machine learning and deep learning models in research-intensive environments. Proven track record of delivering end-to-end data science solutions — from raw data ingestion and statistical modeling to production deployment. Built clinical ICU forecasting pipelines (MIMIC-IV) benchmarking LSTM, GRU, TFT, and Ensemble models, alongside a deployed CNN-based real-time detection web app at ICAR-CIBA. Specializes in predictive modeling, time series forecasting, geospatial analytics, and simulation modeling using Python and R. Experienced in translating complex analytical insights into actionable decisions for domain experts across healthcare and aquaculture domains.

Technical Skills

Languages	Python, R, SQL, LaTeX
ML / DL	Scikit-learn, TensorFlow, Keras, CNN, XGBoost, Random Forest, SVM, LSTM, GRU, Attention-LSTM, TFT, Prophet
Statistics	Regression, Time Series (ARIMA, SARIMA), Simulation, Risk/Vulnerability Assessment, Meta-analysis, DOE
Geospatial	QGIS, Geospatial Analysis, Spatial Statistics
Data & Tools	Pandas, NumPy, Matplotlib, Seaborn, Plotly, SPSS, MS Office
Dev Tools	Git, RStudio, VS Code, Jupyter Notebook, Flask/Web App Integration
Cloud / MLOps	AWS (beginner), Docker (beginner), Model Deployment

Experience

ICAR – Central Institute of Brackishwater Aquaculture (CIBA), Chennai
Senior Research Fellow (Data Scientist – ML & Deep Learning)

Mar 2025 – Present

- Built a **CNN-based image classification model** for real-time potassium ion (K^+) detection integrated into a **live web application**, enabling instant water quality optimization for shrimp culture ponds.
- Developed **XGBoost and Random Forest** ensemble models for shrimp disease prediction using 15+ culture and climate features, improving early-warning detection accuracy over baseline logistic regression.
- Conducted **aquaculture meta-analysis** across shrimp, freshwater, and biofloc systems synthesizing findings from 50+ studies to derive statistically robust productivity benchmarks.

ICAR – Central Institute of Brackishwater Aquaculture (CIBA), Chennai
Project Associate (Data Analyst – Statistical Modeling & Geospatial)

Feb 2022 – Feb 2025

- Designed and deployed an **interactive shrimp growth simulation model** incorporating real-time weather and water quality data, accessible publicly at ciba.res.in/shrimpsim/ and used by aquaculture practitioners nationwide.
- Conducted **risk and vulnerability assessment** for 12+ coastal districts using statistical and GIS-based geospatial techniques, informing policy decisions on climate-resilient aquaculture zones.
- Performed **time series trend analysis** (ARIMA/Mann-Kendall) on 20+ years of climate data, identifying significant seasonal shifts impacting brackishwater aquaculture productivity.

JKKM College of Agricultural Sciences, Erode
Assistant Professor – Statistics & Mathematics

Mar 2021 – Jan 2022

- Delivered undergraduate courses in Agricultural Statistics and Mathematics; mentored 120+ students and served as exam coordinator for semester assessments.

Key Projects

- MIMIC-IV Clinical Time-Series Forecasting Pipeline** | *Python, LSTM, GRU, TFT, Prophet, TensorFlow, Pandas* | github.com/itzkumarc/mimic-iv-forecasting
Built a high-resolution, end-to-end ICU clinical forecasting pipeline on the **MIMIC-IV v2.2 dataset** to predict **Mean Arterial Pressure (MAP)** at a 6-hour horizon. Benchmarked four architectures — **Facebook Prophet** (baseline), **2-layer Stacked Attention-LSTM**, **GRU / Temporal Fusion Transformer (TFT)**, and a **Weighted Ensemble** — across full EDA, QC, training, and evaluation stages. Demonstrates end-to-end ML engineering: raw clinical data ingestion, hourly time-series construction, model training, and automated leaderboard reporting.
- ShrimpSIM** | *R, Rshiny, shinyjs, Simulation Model*
Built an end-to-end simulation tool integrating weather and water quality parameters to predict shrimp growth trajectories. Deployed as a publicly accessible web application used by researchers and farmers.
- CNN-based Real-time K^+ Ion Detection System** | *Python, TensorFlow, CNN, Flask*
Developed a convolutional neural network pipeline for real-time detection of potassium concentration in aquaculture ponds, integrated into a web interface for on-site monitoring.
- Shrimp Disease Prediction using ML** | *Python, XGBoost, Random Forest, Scikit-learn*
Engineered predictive models using culture and climate variables for early detection of white fecal disease in *Penaeus vannamei* shrimp, supporting timely intervention in commercial farms.

- **Coastal Vulnerability Index Mapping** | *R, QGIS, Geospatial Statistics*
Developed a multi-criteria risk and vulnerability index for coastal aquaculture districts using geospatial layers and composite statistical scoring, used for disaster preparedness planning.

Education

Tamil Nadu Agricultural University
M.Sc. Agricultural Statistics (OGPA: 8.5/10.0)
Relevant Coursework: Regression Analysis, Probability Theory, Design of Experiments, Multivariate Statistics, Machine Learning, Statistical Computing

2018 – 2021
Coimbatore

Kumaraguru Institute of Agriculture
B.Sc. Agriculture (OGPA: 8.0/10.0)

2014 – 2018
Erode

Publications

- Statistical evaluation of growth models for area, production, and productivity of tapioca. *International Journal of Agriculture Sciences*
- Spatial and temporal variations of tapioca. *International Journal of Statistics and Applied Mathematics*
- Does abiotic stress influence the white fecal disease incidence and its intensity in *Penaeus vannamei* shrimp farms? *Aquaculture International – Findings integrated into ML disease prediction pipeline.*

Conferences & Presentations

- Machine Learning approach for shrimp disease prediction using culture and climate parameters. *12th Symposium on Diseases in Asian Aquaculture*
- Outlier Detection for Wholesale Price Index of Onion in India. *Emerging Trends in Statistics and Data Science Conference*
- Trends in surface weather parameters of coastal India and implications for brackishwater aquaculture. *XVI Agricultural Science Congress*

Languages

Tamil (Native) — English (Professional Proficiency)